SHRAOSHREE GHOSH

Software Engineer | Backend, Data Systems & Reliability
Thornton, CO (Open to Relocation) | shraoshree27@gmail.com | (520) 204-8107 | GitHub

PROFESSIONAL SUMMARY

Software engineer with 4+ years of experience building and supporting high-throughput backend and data-processing systems using C++, SQL, Python, and PostgreSQL. Improved throughput by 35% for workflows processing 10M+ records daily and sustained 99.9% processing accuracy through validation, reconciliation, duplicate detection, and idempotent reprocessing. Experienced in production reliability, monitoring, CI/CD, REST APIs, and analytics/reporting.

TECHNICAL SKILLS

Programming & Backend: Python, C++, SQL, FastAPI, REST APIs, object-oriented design, data structures and algorithms
Data & Reliability: PostgreSQL, SQL Server, Redis, data modeling, batch processing, validation, reconciliation, duplicate detection, idempotency, caching, Tableau, dbt (data-quality testing)
Production Engineering: Linux, monitoring, logging, health checks, alerting, root-cause analysis, performance tuning, incident response, testing
Cloud & Delivery: AWS (EC2, S3), Docker, Kubernetes, Jenkins CI/CD, Git/Gerrit; Kafka (consumer monitoring, lag troubleshooting, redelivery handling)

PROFESSIONAL EXPERIENCE

TATA CONSULTANCY SERVICES | Client: Nokia | Kolkata, India
Senior Software Engineer | Mar 2023 – Mar 2025

- Optimized C++/SQL backend processing services handling 10M+ telecom records daily; profiled database bottlenecks and implemented batching and caching improvements that increased throughput 35%.
- Supported continuous Kafka-based ingestion for production workflows; troubleshot consumer lag, duplicate messages, application failures, and connectivity issues, using idempotent SQL checks and UPSERTs to safely handle redelivered events.
- Owned 100+ production data workflows across ingestion, validation, transformation, and database writes, sustaining 99.9% processing accuracy.
- Supported production data-quality validation with dbt by modifying and running not-null, uniqueness, and referential-integrity tests; investigated failures through SQL and corrected underlying data and transformation issues.
- Troubleshot and resolved 50+ production incidents across AWS/Kubernetes environments using logs, runtime diagnostics, SQL analysis, and code-level debugging; drove root-cause fixes for reliability and performance issues.
- Performed reconciliation and controlled backfills for missing or failed historical records, using idempotent reprocessing to restore data completeness without creating duplicates.
- Designed technical solutions and supported 10+ releases through Jenkins CI/CD, deployment validation, and rollback procedures; onboarded engineers, led KT sessions, and helped teammates troubleshoot production issues.
- Built scheduled Python automation using cron and Jenkins for production data validation and reporting, detecting discrepancies and reducing manual validation effort by 60% for a 10–12 engineer team.

TATA CONSULTANCY SERVICES | Chennai, India
Software Engineer | Feb 2021 – Feb 2023

- Optimized relational data models and SQL performance by analyzing query patterns and creating/modifying indexes; developed validation and reconciliation logic with duplicate detection and idempotent reprocessing.
- Initiated a Tableau reporting POC that expanded to 7+ dashboards for SLA compliance, MTTR, and ticket-volume insights; traced reporting issues through source tables and records to validate data accuracy.
- Delivered production fixes with unit and integration testing, reducing recurring defects by 25% and improving reliability of data-processing workflows.

PROJECTS

URL Shortener API | Python, FastAPI, PostgreSQL, Redis, Docker
Built a REST API with Python, FastAPI, PostgreSQL, and Redis featuring expiration, click tracking, TTL caching, cache invalidation, rate limiting, Docker Compose, and OpenAPI documentation. github.com/shraoshree/url-shortener

AI Incident Analysis Copilot | Python, FastAPI, Pydantic, Gemini
Developed a Python/FastAPI incident analysis API backed by PostgreSQL/pgvector with structured responses, historical runbook retrieval, retry/fallback handling, health checks, and logging. github.com/shraoshree/ai-incident-copilot

EDUCATION

M.S. Business Analytics, Trine University | Expected May 2027 | GPA 3.67/4.0 (**Able to join immediately**)
M.Tech Computer Science & Engineering, Vellore Institute of Technology | 2020
B.Tech Computer Science, Christian College of Engineering & Technology | 2017